

CHAPTER 1

GENERAL INFORMATION

1.1 INTRODUCTION

This manual provides general information, functional description, maintenance information, and illustrated parts lists for the M68SFDC1000/1010/2000 EXORdisk II Disk Drive Unit utilizing two CalComp 142M disk drives. The Disk Drive Unit is shown in Figure 1-1. All address references within this manual are shown in hexadecimal unless otherwise indicated.

1.2 FEATURES

The features of the Disk Drive Unit include:

- Two floppy disk drive units capable of storing 595,888 bytes of IBM 3740 formatted data (292,944 bytes per drive unit)
- Fast access. Track-to-track access time of 6 milliseconds with head stabilization time of 10 milliseconds.
- Individually selectable disk drives
- Both disk drives, the power supply, and a cooling fan are contained in a single housing
- All interface and control is provided by a single module contained within the EXORciser or microcomputer system (the M68SFDC Floppy Disk Controller Module)

1.3 SPECIFICATIONS

The specifications for the Disk Drive Unit are identified in Table 1-1.

1.4 GENERAL DESCRIPTION

The Disk Drive Unit provides the two disk drives and power supply for the EXORdisk II Floppy Disk System, all mounted within a single tabletop housing. The power supply provides the +24 Vdc power required to operate the disk drives. The +5 Vdc power for the disk drive logic is provided by the host system (i.e., EXORciser) via the interconnecting cable assembly. Drive control and interface is provided by the M68SFDC Floppy Disk Controller Module, which is installed in either the EXORciser or microcomputer system. Interconnection between the Disk Drive Unit and the Floppy Disk Controller Module is provided by a 40 conductor flat ribbon cable. (Refer to the M68SFDC Floppy Disk Controller Module manual for information concerning the controller module.)

The two CalComp model 142M disk drives used within the Disk Drive Unit are high-speed, random-access storage units, utilizing a flexible disk cartridge as the storage medium. Each drive features 48 track-per-inch and 6400 bits-per-inch technologies. Access time is 6 milliseconds track-to-track with 10 milliseconds required after the final step for head stabilization. The flexible disk is rotated at 360 rpm, yielding a data transfer rate of 500,000 bytes-per-second. Both disk drives are mounted in the housing side-by-side and connected in daisy-chain fashion.

TABLE 1-1. Disk Drive Unit Specifications

CHARACTERISTICS	SPECIFICATIONS
Power Requirements	
AC Power	115/230, 50/60 Hz, 350 Watts (Total)
M68SFDU1000 (Each Drive)	115 Vac $\pm$ 10%, 60 $\pm$ 0.5 Hz @ 1.7 Amps (Max)
M68SFDU1010 (Each Drive)	115 Vac $\pm$ 10%, 50 $\pm$ 0.5 Hz @ 1.7 Amps (Max)
M68SFDU2000 (Each Drive)	230 Vac $\pm$ 10%, 50 $\pm$ 0.5 Hz @ 0.9 Amps (Max)
DC Power (Each Drive) (Provided by Internal Supply)	+24 Vdc $\pm$ 5% @ 1.5 Amps (Max)
Disk Drives (Each)	
Storage Capacity (IBM 3740 Format)	
Per Flexible Disk	292,944 bytes (1,943,552 bits)
Per Track	3,328 bytes (26,644 bits)
Per Sector	128 bytes (1,024 bits)
Data Transfer Rate	250 Kbits/sec. (2.4 ms/sector)
Access Time	6 ms track-to-track
Head Stabilization Time	10 ms
Head Load Time	30 ms
Rotational Speed	360 $\pm$ 2.5%
Latency	166.7 ms (Max) 83.3 ms (Avg)
Recording Method	Frequency Modulation
Operating Temperature	10° to 37.8° C. (50° to 100° F.)
Physical Characteristics	8.4 in. (W) x 15.75 in. (D) x 4.9 in. (H)
Connector Types (Located on Drive)	
AC Power (3 Pin) (J0)	Molex 1396 (Molex 1396R Mating)
DC Power (4 Pin) (J2)	AMP 350211-1 (AMP 1-480424-0 Mating)
Disk Drive (40 Pin) (J1)	3M 3432 (3M 3417-3000 Mating)
Drive Unit	
Physical Dimensions	17.75 in. (W) x 23.5 in. (D) x 6.96 in. (H)
Weight	48 Pounds

1.5 RELIABILITY

1.5.1 Service Life

The disk drives used within the EXORDisk II (CalComp 142M) are designed and constructed to provide a useful life of 5 years or 30,000 hours, whichever is less, before factory overhaul or replacement is required. This does not include the repair or replacement of parts that may be required during the lifetime of the unit.

1.5.2 Preventive Maintenance

No regular preventive maintenance program is required. For those desiring to perform periodic checks, a suggested schedule is provided in the Maintenance Section of this manual.

1.5.3 Mean Time Between Failures (MTBF)

Following an initial period of 200 hours of operation, the MTBF of each disk drive is expected to exceed 5,000 hours. MTBF is defined as:

$$\text{MTBF} = \frac{\text{Operating Hours}}{\text{Number of Equipment Failures}}$$

Operating hours means total *power-on* hours less maintenance time (if any). Equipment failures mean any stoppage or substandard performance of the unit because of equipment malfunction. Equipment failure excludes downtime or substandard performance due to operator error, adverse environment, power failure, or other failures not caused by the drive. To establish a meaningful MTBF, operating hours must be greater than 3,500 hours and includes an average of all drives in use. Equipment failures are defined as those actions requiring repairs, adjustments, or replacements on an unscheduled basis.

1.5.4 Mean Time To Repair (MTTR)

The MTTR of the disk drive is less than 0.5 hour, and is defined as the time required for an adequately trained and competent serviceman to diagnose and correct a malfunction.

1.5.5 Recoverable Read Error Rate

The recoverable read error rate is less than one error in 10^9 bits read. A recoverable read error is one that can be recovered in ten or fewer attempts to read the record.

Errors attributed to the flexible diskette are not included in determining the recoverable read error rate.

1.5.6 Non-Recoverable Read Error Rate

The non-recoverable read error rate is less than one error in 10^{12} bits read. A non-recoverable read error is one that remains after ten unsuccessful attempts to read the record.

Errors attributed to the flexible diskette are not included in determining the non-recoverable read error rate.

1.5.7 Access Positioning Error Rate

Providing that the M68SFDC Floppy Disk Controller Module is performing correctly, the access positioning error rate is less than one error per 10^6 positioning operations.

1.5.8 Read/Write Error Detection

Write errors are usually detected as read errors making it difficult to detect whether the read or write operation was in error. A Mini-Diagnostic Disk firmware test routine is incorporated in the M68SFDC Floppy Disk Controller Module that permits sectors on the floppy diskette to be individually written to or read from, thus enabling read or write errors to be identified. The procedure for performing read and write tests is provided in the Testing paragraph (refer to the Maintenance Section).

1.6 WRITE PROTECTION

Write protection is provided as a means of protecting against inadvertent destruction of data. The write protect feature uses a phototransistor to sense the presence or absence of a write-protected notch in the edge of the flexible disk (refer to Figure 1-2). Diskettes may be purchased with or without the write-protect notch. If purchased without the notch, then a notch may be cut into the diskette envelope. The necessary location and dimension information to cut this notch is also provided in Figure 1-2.

When write protection is used, any disk lacking an opaque write-enable sticker is write-protected. This means that the drive cannot record data on the diskette.

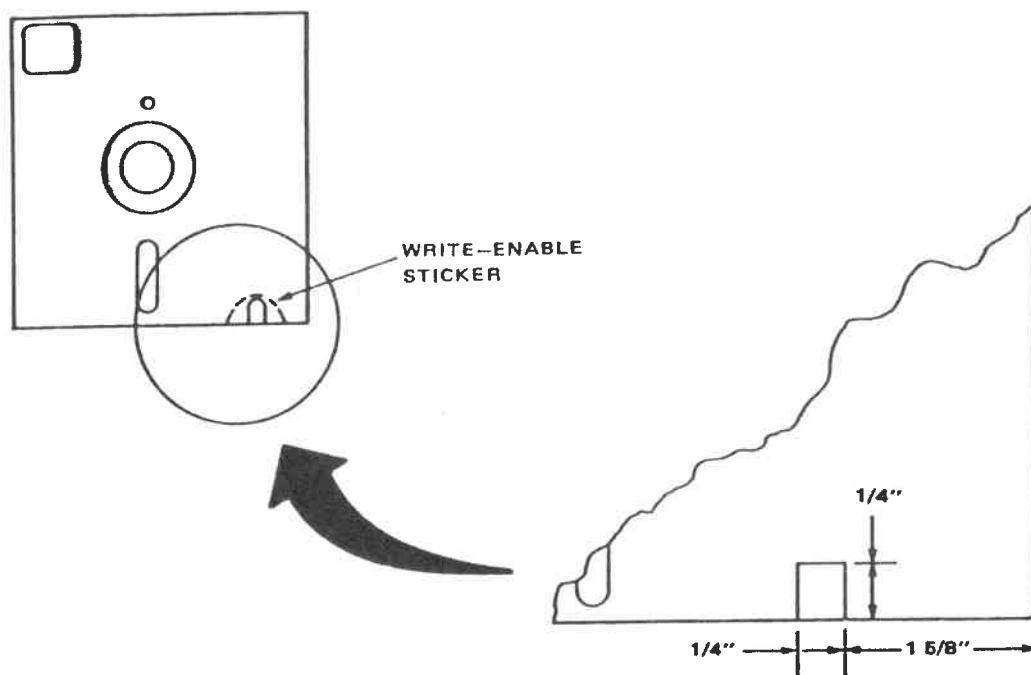


FIGURE 1-2. Write Protection

CHAPTER 2

FUNCTIONAL DESCRIPTION

2.1 INTRODUCTION

This chapter provides a functional description of the EXORDisk II Disk Drive Unit. Since this Disk Drive Unit consists mainly of two CalComp Model 142M disk drives and a power supply, this functional description is limited to a discussion of the disk drive only.

2.2 DESCRIPTION

The Model 142M disk drive contains a single data and control board, the flexible disk drive mechanism, the read/write head positioning mechanics, and the necessary transducers. A block diagram of the disk drive is provided in Figure 2-1. The elements that comprise the disk drive combine to accomplish the following:

- Interpret input interface signals
- Position the read/write head to the desired track
- Read data from and write data to the flexible diskette
- Provide status information to the M68SFDC Floppy Disk Controller Module

The data and control printed circuit board contains the following circuits:

- Input interface receivers
- Stepper motor control and drive circuits
- Index detection logic
- Sector detection logic
- Write-protection detection logic
- Track 00 detection logic
- Head load actuator control and drive circuit
- Read circuit
- Digital Time Delay Filter (DTDF)
- Data separator circuit
- Write circuit
- Output interface drivers

2.2.1 Spindle Drive

The flexible disk is driven at 360 rpm by a spindle that is attached by a drive belt to an ac drive motor. A registration hub that precisely registers the flexible disk by the use of an expansion centering cone is located on the chassis. As the cover is closed, the centering cone expands to precisely register the flexible disk on the drive spindle.

2.2.2 Track Accessing

The read/write head is moved to the correct track in response to the DIRECTION and STEP input interface lines. Each step pulse causes the stepper motor to rotate in 30° increments clockwise or counterclockwise as defined by the state of the direction line. Since the head is mounted on a carriage that is driven by the helical shaft of the stepper motor, each 30° of rotational movement is converted to 0.021 inch of linear movement (one track).

As the last step is issued, the media is located against the head by the HEAD LOAD interface line in preparation for data transfer. The time required to move the last track is 6 ms; 10 ms is required for mechanically stabilizing the head; media loading time is 30 ms. Thus, by loading the media simultaneous with issuing the last step pulse, the first 16 ms of media loading time occurs during the time required for stepping the last track and for mechanical stabilization of the head. This results in higher throughput.

2.2.3 Read/Write Head

The Model 142M uses a single element, double-density, ceramic read/write head that is designed to provide maximum signal playback with minimal wear. The head also contains a tunnel-erase coil to erase areas between data tracks to compensate for normal tolerances between flexible disks and disk drives. These erased areas ensure media interchangeability as well as high data integrity.

The read/write head is mounted on a carriage that is driven by the helical shaft of the stepper motor. The head is held in a plane precisely perpendicular to the flexible disk to ensure perfect head-to-disk compliance. The flexible disk is loaded against the head in response to the HEAD LOAD input interface signal (See Figure 2-2).

2.2.4 Flexible Diskette

The recording medium specified for use with the Model 142M is an oxide-coated, flexible disk enclosed in a protective plastic envelope. The plastic envelope contains apertures for spindle loading, head contact, and sector/index detection.

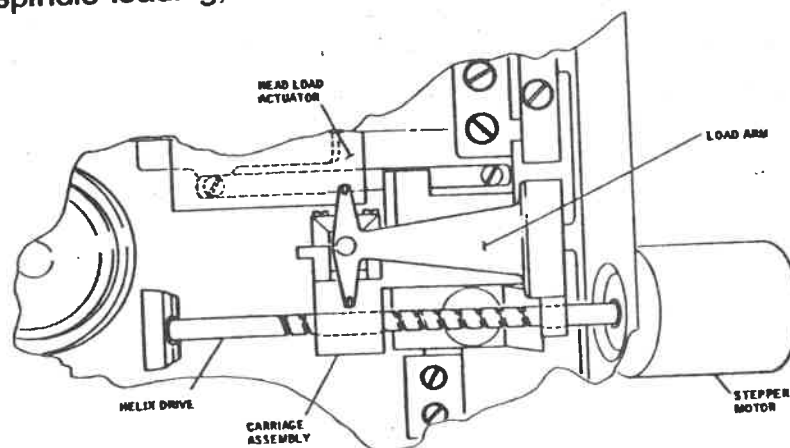


FIGURE 2-2. Carriage Assembly

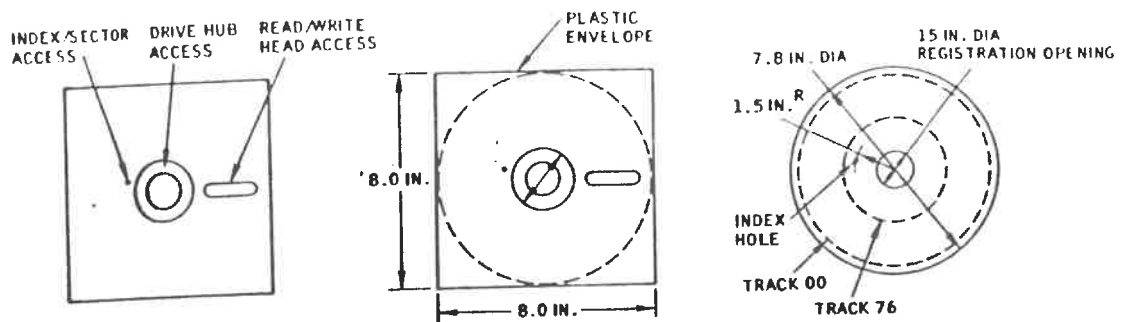


FIGURE 2-3. Flexible Disk Construction

2.2.4.1 Diskette Parameters. The recommended flexible disk is the IBM 3740 Diskette or equivalent (refer to Figure 2-3). Flexible disk characteristics are as follows:

• Disk Diameter	7.875 inches
• Envelope Size	8 x 8 inches
• Rotational Speed	360 rpm $\pm 2.5\%$
• Rotational Period	166.67 ms $\pm 2.5\%$
• Average Latency	83.33 ms
• Tracks	77
• Track Density	48 tpi
• Bit Density	6536 bpi (Inside Track)

2.2.4.2 Diskette Environment. Flexible disks must be in the same temperature and humidity environment as the disk drive for a minimum of 5 minutes before installing the flexible disk in the disk drive. These environmental requirements are as follows:

- Temperature 50°F (10°C) to 125°F (51.6°C); maximum gradient of 20°F (11.1°C) per hour
- Relative Humidity 8% to 80%, maximum wet bulb 85°F (29.4°C)

2.3 TRACK ACCESSING DESCRIPTION

Track accessing is accomplished by using the DIRECTION and STEP interface lines. The DIRECTION line is set to the proper level and is stable for 100 nanoseconds, minimum, before the leading edge of STEP. Figure 2-4 depicts track accessing timing.

Head positioning is accomplished by pulsing the STEP line at 167 pulses per second with one resultant head position change per pulse. Ten milliseconds of head stabilization time is allowed after the last step prior to proceeding with data transfer.

NOTE

The disk drives are designed to operate at a step pulse rate of 167 pulses per second. Operation at lower step pulse rates may result in slightly noisy operation.

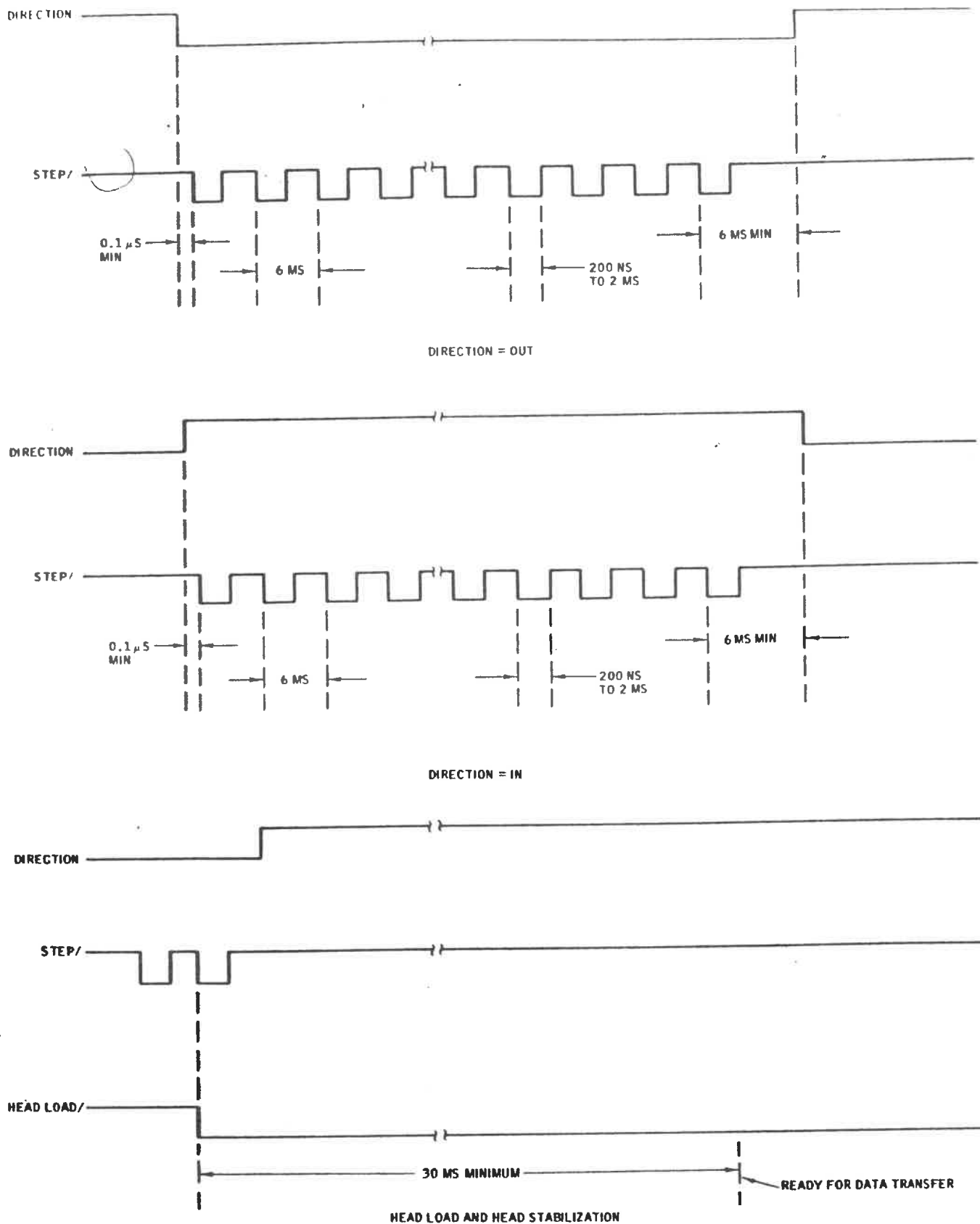


FIGURE 2-4. Track Accessing Timing Diagram

If a data transfer is to commence upon locating the desired track, the head load operation overlaps the seek operation. By using this technique, the time to validate data is reduced.

Should an access position error occur, a reference may be established by either reading the track header or by stepping the head out until the TRACK 00 line becomes active.

2.4 READ DATA TIMING DESCRIPTION

Read data is transferred in a serial fashion between the M68SFDC Floppy Disk Controller Module and the disk drive. For clarity, the read data output is depicted separately along with appropriate timing details in Figure 2-5.

2.5 WRITE DATA TIMING DESCRIPTION

Write data is transferred in a serial fashion between the M68SFDC Floppy Disk Controller Module and the disk drive. The WRITE ENABLE interface line controls both erase and write amplifiers. The erase amplifier is enabled at the leading edge of WRITE ENABLE and is disabled 600 microseconds after the fall of WRITE ENABLE. This delay is provided to ensure final trimming of the just-written data. See Figure 2-6 for write timing waveforms.

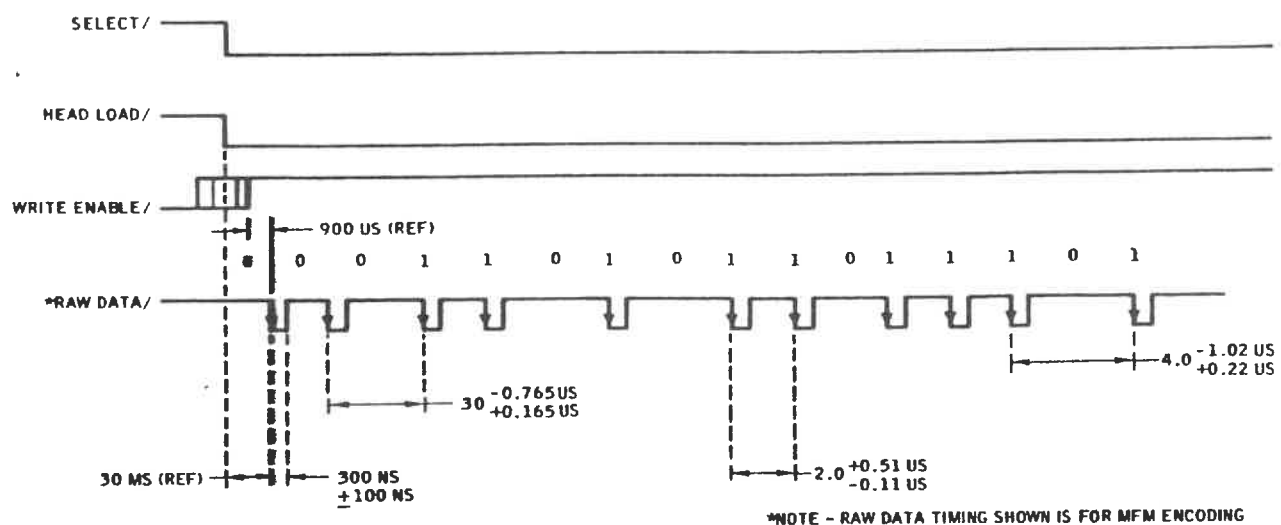


FIGURE 2-5. Raw Data MFM Code Read Signal Timing Diagram

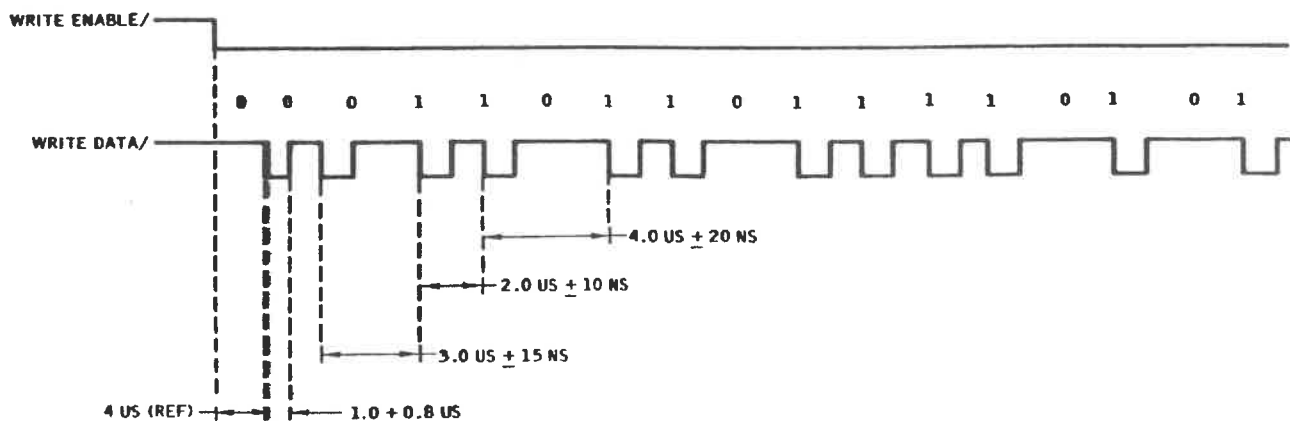


FIGURE 2-6. Write Data Timing Diagram

2.6 POWER-ON TO DATA TRANSFER DESCRIPTION

Power-on to read timing and to write timing is depicted by Figure 2-7. In both cases, it is assumed that the controller will step the head to Track 00 for a reference before stepping to a specified track. It is not necessary to have a diskette loaded to step the read/write head. However, for purposes of explanation, it is assumed that a diskette is loaded and that the controller is waiting for the READY interrupt before initiating head stepping.

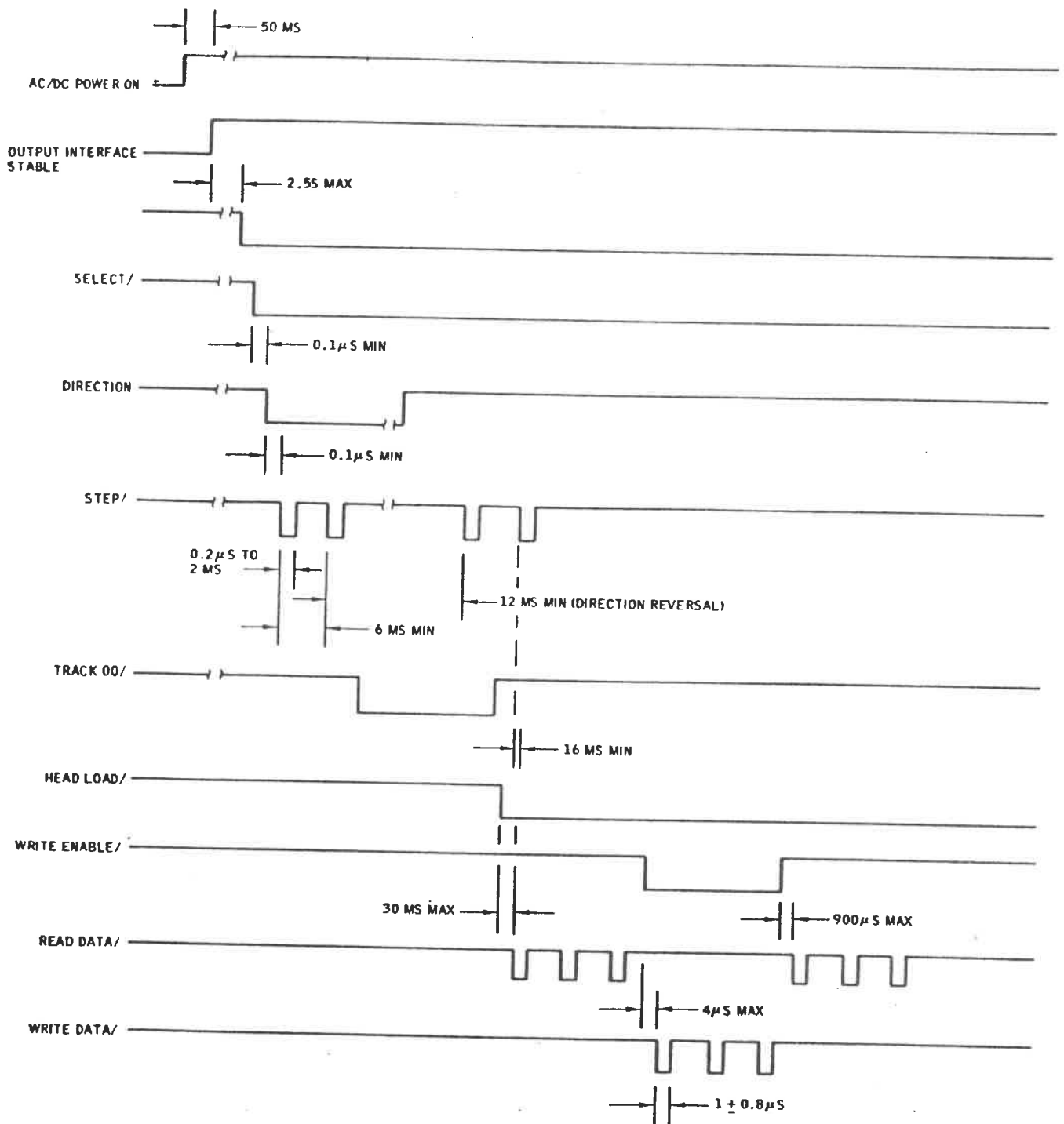


FIGURE 2-7. Power On, Positioning, Read, and Write Timing Diagram

CHAPTER 3

MAINTENANCE

3.1 CLEANING INSTRUCTIONS

The ~~EXOR~~ disk II does not normally require periodic cleaning of the chassis nor does the disk ~~ive~~ read/write heads. Read/Write head contamination is normally prevented by a special ~~wiping~~ fabric contained within the protective envelope of each flexible diskette. This wiping ~~fabric~~ removes and traps excessive oxide or other foreign matter from the surface of the disk ~~ette~~. However, should the read/write head require cleaning due to the continued use of a ~~dirty~~ contaminated flexible diskette, the head may be cleaned using lintless gauze dampened ~~with~~ isopropyl alcohol. After cleaning, the head should be wiped with a clean, dry gauze ~~to~~ remove any residue left when the alcohol evaporates.

A ~~tan~~ filter element, installed in the rear of the Disk Drive Unit, should be cleaned whenever ~~the~~ condition of the filter indicates cleaning is required. This depends greatly on the operating environment of the Disk Drive Unit. To clean this filter, remove it and rinse in clean, ~~fresh~~ water. Allow the filter to dry thoroughly before reinstalling it in the chassis.

3.2 PREVENTIVE MAINTENANCE

The following maintenance schedule is suggested for those who wish to perform periodic ~~checks~~:

Monthly

1. ~~Clean~~ all exterior surfaces with a cloth dampened with a nonabrasive cleaning solution.
2. ~~Examine~~ the air filter and clean the element as necessary.

Semiannually

1. ~~Clean~~ the interior surfaces of both disk drives, drive belts, drive pulleys, and carriage assemblies with alcohol and lint-free cloths.
2. ~~Visually~~ examine the read/write head for scratches, wear, and oxide deposits. If the head ~~is~~ dirty, clean it with isopropyl alcohol and remove any residue with a lint-free cloth. **DO NOT** touch the surface of the head. If the head has a defective surface, replace the head carriage assembly.
3. ~~Perform~~ a read/write test on each disk drive to verify proper operation (refer to the ~~Testing~~ paragraph).

3.3 TESTING

The best method of testing EXORdisk II operation is to write 128 bytes of data from memory onto one sector of a floppy diskette and then read the data from that sector into a second memory block of 128 bytes. This permits the data in the two memory blocks to be compared in order to verify the disk read and write operation. The firmware incorporated in the M68SFDC Floppy Disk Controller Module includes a disk diagnostics routine that permits individually selected sectors on the floppy diskette to be written to and read from. If the EXORdisk II is tested using an EXORciser, the following test procedures may be used directly. However, if the EXORdisk II must be tested using any other microcomputer-based system (such as a Micromodule-based system), then the following procedures must be modified for that system. For this reason, the disk parameters that must be established prior to test execution (both the memory locations and the data stored at those locations) are provided in Table 3-1.

The test procedure is divided into two sections: Write Test and Read Test. These two procedures should be performed in the sequence presented to completely evaluate system operation.

3.3.1 Write Test Procedure

1. Initialize the EXORdisk II and EXORciser system and insert a new floppy diskette into the drive to be tested.
2. Enter the EXbug's MAID routine.
3. Enter 128 bytes of data starting at address \$0100.
4. Enter the number of the drive to be tested (0 or 1) at address location \$0000.
5. Enter the number of the disk sector (in hex) onto which data is to be recorded (sectors \$0000 to \$07D1) at address locations \$0001 (MSB) and \$0002 (LSB). *Suggestion:* use the first sector \$0000. This enables the last sector on the diskette (\$07D1) to be used in a later step, permitting operation to be tested below and above track 43. (Above track 43, space compression is used for all recorded data, since the length of each track is effectively shorter than the tracks below track 43.)
6. Enter the number of contiguous sectors (in hex) to be written onto (in this case only 1 — \$0001) at address locations \$0003 (MSB) and \$0004 (LSB).
7. Enter the number \$80 at address location \$0005.
8. Enter the starting address of the data to be written onto the diskette (in this case \$0100) at address locations \$0020 (MSB) and \$0021 (LSB).

TABLE 3-1. Parameters Required To Test Read/Write Disk Operation

ADDRESS LOCATION	NAME	CONTENT SET BY USER
\$0000	CURDRV	Disk drive number (0 or 1)
\$0001 (MSB) \$0002 (LSB)	STRSCT	Starting sector number of first sector to be written onto or read from (sectors range from \$0000 to \$07D1)
\$0003 (MSB) \$0004 (LSB)	NUMSCT	Number of contiguous sectors to be written onto or read from
\$0005	LSCTLN	Length of last sector to be loaded (enter \$80)
\$0020 (MSB) \$0021 (LSB)	LDADDR	Starting memory address of the buffer from which data is to be written from or read into
\$0022 (MSB) \$0023 (LSB)	EXADDR	The address of the Read routine (\$E869) or Write routine (\$E884)
\$0024	ONECON	Set to any non-zero value

9. Enter the address of the Write routine (\$E884) at address locations \$0022 (MSB) and \$0023 (LSB).
10. Enter any non-zero number at address location \$0024.
11. Enter the command: EB98;G to invoke the write routine. After all 128 bytes (1 sector) of data have been written from memory onto the selected disk sector, a disk completion status message (E0 through E9) will be returned and control will be returned to EXbug. If an E0 status message is returned, no errors have occurred and the data has been written onto the indicated sector. This can be verified by performing the following Read Test Procedure. If any other status message is returned (E1 through E9) an error has occurred during the write operation. Table 3-2 explains each of these error messages.

3.3.2 Read Test Procedure

This procedure is intended to be performed after execution of a Write Test Procedure.

1. Enter the EXbug's MAID routine.
2. Enter the starting address in memory at which the data from the diskette will be read into (such as \$0200) at address locations \$0020 (MSB) and \$0021 (LSB).

TABLE 3-2. Error Messages

MESSAGE	EXPLANATION
E0	NO ERROR — Normal response
E1	DATA CRC ERROR — The CRC information following the sector did not match the calculated result.
E2	WRITE PROTECTED DISK — The write-protection tab on the diskette has not been covered (refer to paragraph 1.6).
E3	DISK NOT READY — The diskette is not turning at full speed. Could be caused by a drive door being left open or a worn drive belt.
E4	DELETED DATA MARK — An attempt was made to read a sector prefaced with a deleted data mark.
E5	TIMEOUT — The timer on the M68SFDC Floppy Disk Controller Module has timed out before the disk operation was completed. In addition, any NMI instruction (such as depressing the ABORT button on the EXORciser prior to completion of the disk operation) will cause this error message to be returned.
E6	INVALID DISK ADDRESS — An attempt was made to access a sector that lies beyond the end of the disk. This error message will also be returned if, during execution of a multiple sector access request, the disk boundary were to be exceeded (such as requesting 10 sectors to be read from the disk when 10 more sectors are not on the disk).
E7	SEEK ERROR — One of the following: 1. A track 0 indication was sensed before the seek operation was completed. 2. A restore function was completed incorrectly. 3. Drive not connected to Floppy Disk Controller Module.
E8	DATA MARK ERROR — A valid data mark could not be found for the specified sector.
E9	ADDRESS MARK CRC ERROR — The CRC information for the address mark does not match the calculated result.

3. Enter the address of the Read routine (\$E869) at address locations \$0022 (MSB) and \$0023 (LSB).
4. Enter the command: EB98;G to invoke the read operation. After all 128 bytes of data have been read from the sector into the selected memory location (in this case starting at address \$0200), a disk completion status message (E0 through E9) will be returned and control will be returned to EXbug. If an E0 status message is returned, no errors have occurred and the data at address locations \$0100 through \$017F (original data written to diskette) can be compared with the data at address locations \$0200 through \$027F (data read from diskette). If any other status message is returned following the read operation (E1 through E9) an error has occurred during read. Table 3-2 explains each of these error messages.
5. If no errors have been detected during the preceding read/write operations, enter the number of the last sector on the diskette (\$07D1) at address locations \$0001 (MSB) and \$0002 (LSB). Then, starting at step 6 of the Write Test Procedure, perform a complete write and read operation. This will verify that these operations perform at both the beginning and end of the diskette.

3.4 POWER SUPPLY ADJUSTMENT

The EXORdisk II Disk Drive Unit incorporates a single +24 Vdc power supply within the chassis (the +5 Vdc power is provided from the host system via the interconnect cable assembly). Therefore, only the +24 Vdc adjustment is provided in the Disk Drive Unit. If the +5 Vdc power supply requires adjustment, refer to the appropriate documentation for the host system.

Potentiometer R11 on the +24 Vdc power supply module (refer to Figure 3-1) is used to adjust the output voltage of the supply. Connect the leads of a multimeter to the terminal posts shown in Figure 3-1 and adjust R11 to achieve +24 Vdc $\pm$ 5%.

3.5 DISK DRIVE MAINTENANCE AND ADJUSTMENTS

The following paragraphs provide maintenance and adjustment procedures for the disk drive.

3.5.1 Belt Tension for New Belts

Initial adjustment for new drive belts must provide between 4 and 5 pounds of tension and later checked to ensure that the tension adjustment has not settled below 2 pounds.

To adjust drive belt tension, proceed as follows:

1. Turn off ac power
2. Remove the disk drive from the chassis

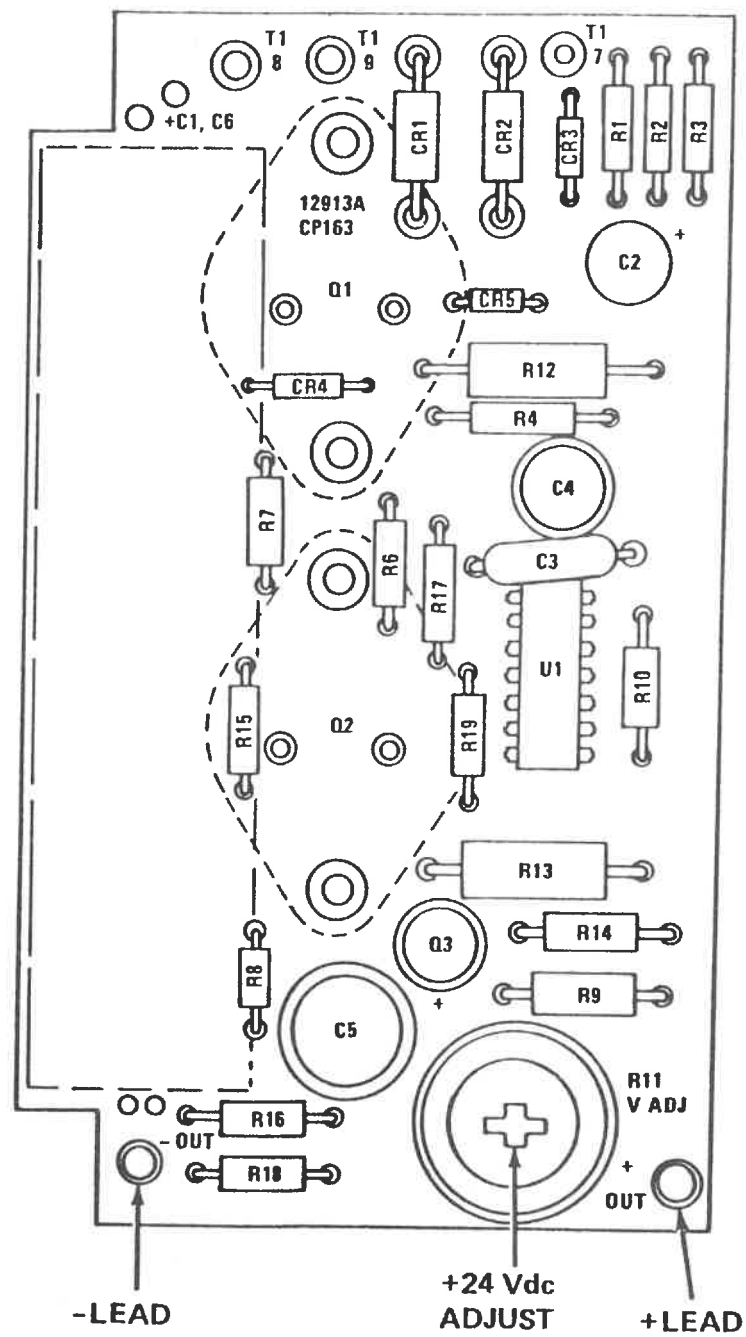


FIGURE 3-1. EXORdisk II Power Supply Test and Adjustment Points

3. Loosen the four drive motor mounting nuts shown in Figure 3-2.
4. Using a spring gauge, move the motor until belt tension is 2.5 to 3.0 pounds.
5. Tighten the four drive motor mounting nuts to maintain the tension between 2.5 and 3.0 pounds.

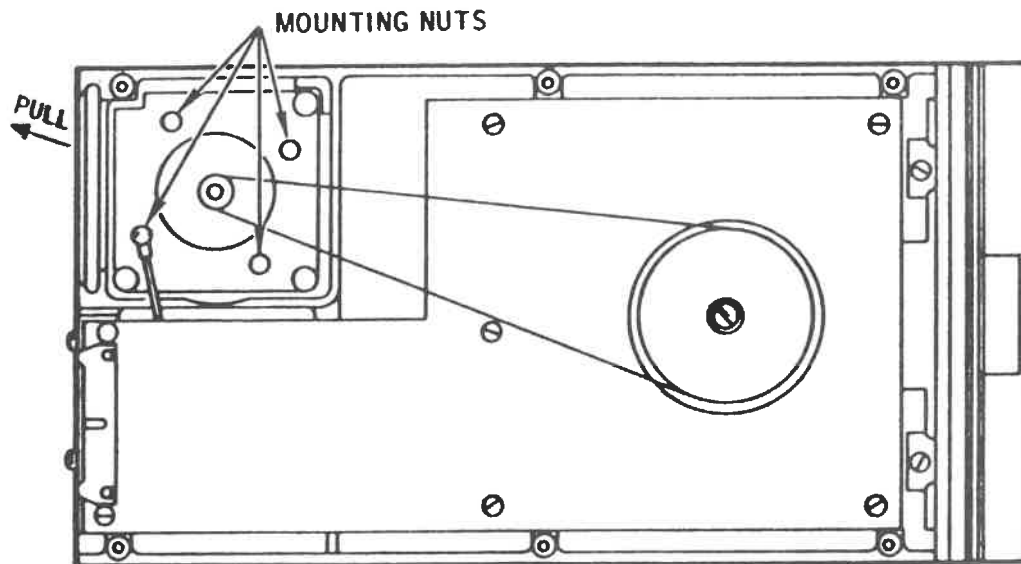


FIGURE 3-2. Drive Belt Check and Adjustment Points

3.5.2 Head Alignment

1. Insert Alignment Diskette (CalComp P/N 17037-001) into the disk drive.

2. Set up the oscilloscope as follows:
 - SYNC: EXT NEG
20 ms/div
Connect probe to A16 pin 2 (Test Point B; INDEX; Figure 3-3).
 - CH A: AC
100 mv/div
Connect probe to junction A18-R73 (Test Point E; DATA; Figure 3-3).
 - CH B: AC
100 mv/div
Connect probe to junction A18-R72 (Test Point F; DATA; Figure 3-3).
 - MODE: Add
Channel B inverted
3. Step carriage to Track 38.
4. The two lobes of the cat-eyes display should be adjusted to obtain equal amplitude as shown in Figure 3-4.
5. To adjust the display, loosen the stepper motor mounting screws and rotate the motor clockwise or counterclockwise to obtain minimum difference in lobe amplitude.

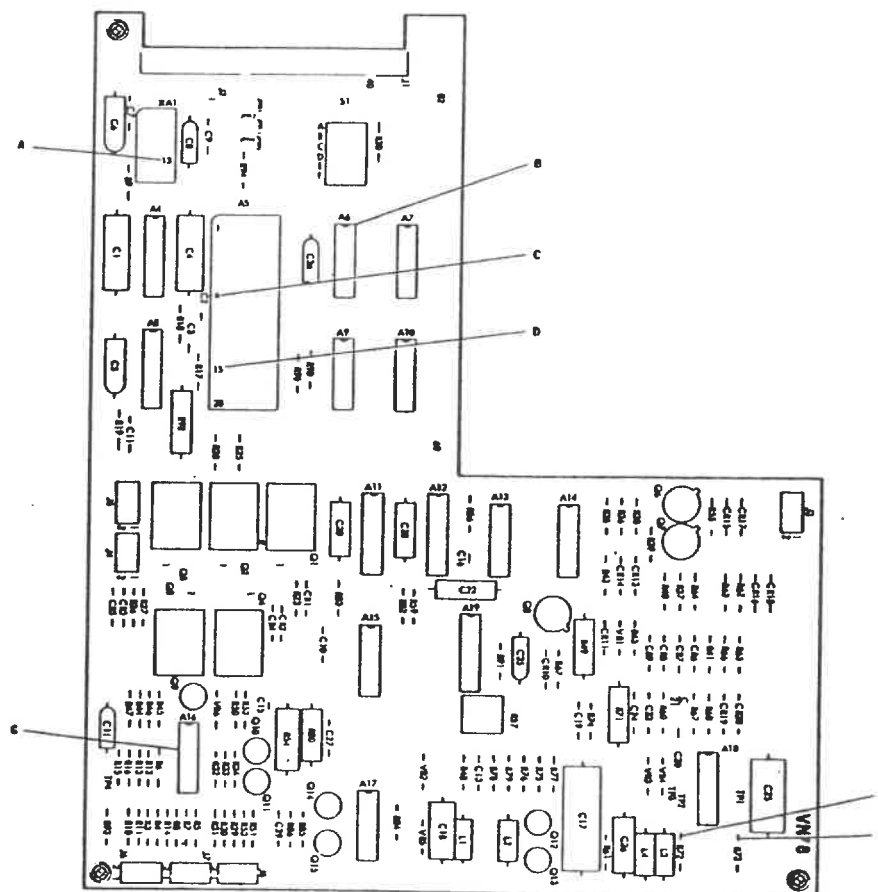


FIGURE 3-3. Data and Control Board Test Points

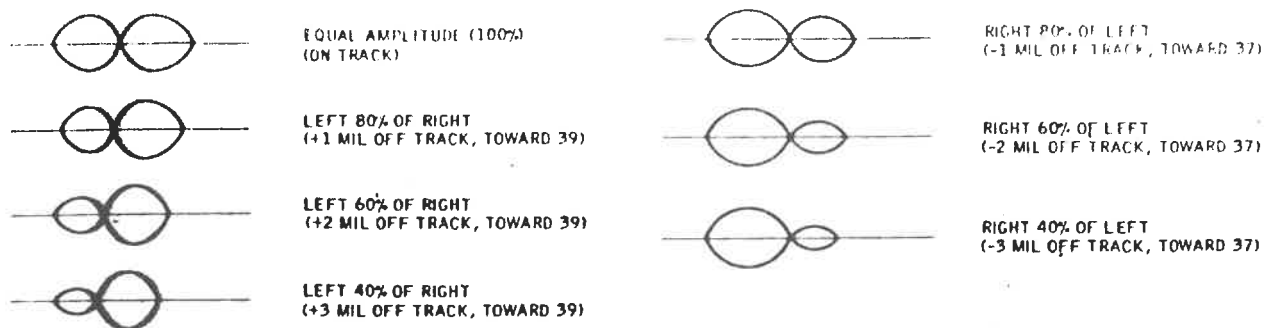


FIGURE 3-4. Cat-Eyes Waveforms

6. Tighten the mounting screws.
7. Step the carriage OUT to Track 00 then IN to Track 38 and repeat steps 5 and 6 as necessary.
8. Step carriage IN to Track 76 then OUT to Track 38 and repeat steps 5 thru 8 as necessary.
9. Remove the alignment diskette; then turn power off.

3.5.3 Track 00 Alignment

Perform this alignment in conjunction with the Head Alignment procedure.

1. Insert Alignment Diskette (CalComp P/N 17037-001).
2. Check for cat-eye lobes as described in the Head Alignment procedure of paragraph 3.5.2.
3. Set up the oscilloscope as follows:
 - SYNC:** INT NEG
20 ms/div
 - CH A:** DC
1v/div
Connect probe to A5 pin 15 (stepper phase 1, Test Point D, Figure 3-3).
4. Step the carriage to Track 38 and verify that the signal is logic level 1 or high, and proceed to step 6. If not, proceed to step 5.
5. If the signal is logic level 0 or low, step IN to Track 39 and verify that stepper phase 1 is high. Perform head alignment as described in the Head Alignment procedure.
6. Step OUT to Track 00.
7. Set up the oscilloscope as follows:
 - SYNC:** INT NEG
1 ms/div
 - CH A:** DC
2v/div
Connect probe to XA1 pin 13 (Step Out) (Test Point A, Figure 4-2)
 - CH B:** DC
2v/div
Connect probe to A16 pin 1 (Track 00) (Test Point G, Figure 4-2)
 - MODE:** Chopped

8. Alternately single step from Track 00 to Track 1.
9. Verify that Track 0 is active between 4.0 ms and 5.0 ms after the step out pulse as shown in Figure 3-5.
10. To make an adjustment, loosen the Track 00 transducer. Adjust with only slight movement of the transducer block.
11. Tighten the screws and reverify by repeating steps 8, 9 and 10 until complete.

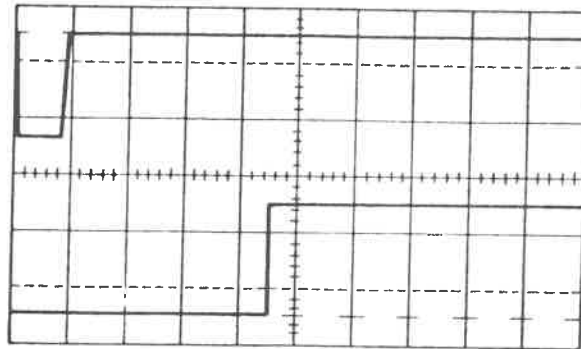


FIGURE 3-5. Track 00 Alignment Waveforms

3.5.4 Index to Data Alignment

1. Insert Alignment Diskette (CalComp P/N 17037-001).
2. Set the oscilloscope as follows:
 - SYNC: EXT NEG
50 us/div
Connect external sync probe to A16 pin 2 (Test Point B, INDEX, Figure 3-3)
 - CH A: AC
200 mv/div
Connect probe to junction A18-R72 (Test Point F, DATA, Figure 3-3)
 - CH B: AC
200 mv/div
Connect probe to junction A18-R73 (Test Point E, DATA, Figure 3-3)
 - MODE: Add
Channel B inverted
3. Step carriage to Track 1.
4. The data burst display should be adjusted as shown in Figure 3-6.

NOTE

Leading edge of data burst is 200 ± 100 us from start of sweep.

5. Step the carriage to Track 1.
6. Loosen the holding screw on the transducer detector (on the chassis near the front panel) until the transducer is just able to move.
7. Observe the timing; adjust the transducer until the data burst is 200 ± 50 microseconds

8. Tighten the holding screw on the transducer detector.
9. ~~Re~~check timing and verify 200 ± 100 microseconds from index data burst on Tracks 1 and 76.
10. ~~Ad~~just the transducer on the cone lift cover for maximum amplitude.
11. ~~Re~~move the alignment diskette and turn power off.

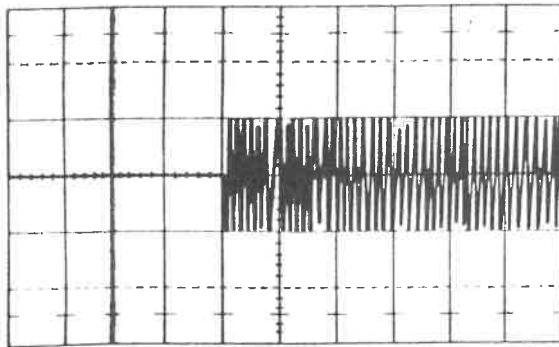


FIGURE 3-6. Data Burst Waveform

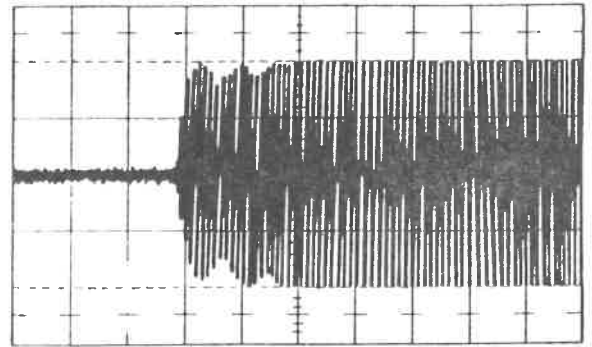


FIGURE 3-7. Head Load Waveform

3.5.5 Head Load Timing

1. ~~Inst~~all a prerecorded diskette with Track 38 fully written.
2. ~~Set~~up the oscilloscope as follows:
 - SYNC:** INT NEG
5 ms/div
 - CHA:** DC
2v/div
Connect scope probe to A5 pin 6 (Test Point C, HEAD LOAD, Figure 3-3)
 - CHB:** DC
200 mv/div
Connect scope probe to junction A18-R73 (Test Point E, DATA, Figure 3-3)
 - MODE:** Channel B
3. ~~Sl~~op the carriage to Track 38.
4. ~~Ob~~serve Channel B; toggle head load, and sync the scope to observe the trace.
5. ~~En~~sure that the data trace reaches 90% of maximum amplitude in 30 milliseconds or less (Figure 3-7).
6. ~~To~~adjust, remove access cover to cone lift cover assembly.
7. ~~Turn~~ the adjusting screw (located on the cone lift cover above the solenoid) clockwise (to reduce time) or counterclockwise (to increase time) until the required result is obtained.

3.5.6 Head Settling Time

1. ~~Insert~~ a scratch diskette with data recorded on Track 1 and data recorded on Track 00.
2. ~~Set~~up the oscilloscope as follows:
 - SYNC:** INT NEG
5 ms/div

CH A: DC
2v/div
Connect probe to XA1 pin 13 (Test Point A, STEP, Figure 3-3)

CH B: AC
100 mv/div
Connect probe to junction A18-R72 (Test Point F, DATA, Figure 3-3)

MODE: Channel B only

3. Alternately single step the carriage between Track 00 and Track 1.
4. Observe that data on Channel B reaches 90% of full amplitude in less than 16 milliseconds at Track 1 (Figure 3-8).

NOTE

Figure 3-9 is the sync (Step) pulse on Channel A (reference only).

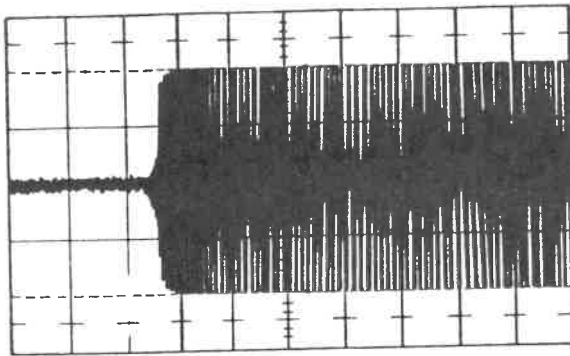


FIGURE 3-8. Head Settling Waveform

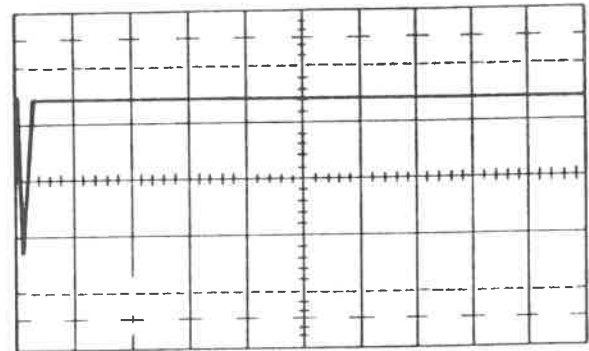


FIGURE 3-9. Step Pulse Waveform

3.6 DISK DRIVE SWITCHES

The Model 142M disk drive incorporates a six-section, SPST, Dual Inline Package (DIP) switch and a jumper plug to provide operational flexibility. Both the DIP switch and the jumper plug are located on the Data and Control Board near the interface connector. The proper settings for the DIP switch are shown in Table 3-3, while the proper connection for the jumper plug is shown in Table 3-4.

TABLE 3-3. DIP Switch Settings

DIP SWITCH	POSITION		FUNCTION
	OPEN	CLOSED	
1 2	X —	— X	Applies raw data to connector J1, pin 38
3	X	—	WRITE ENABLE signal controlled by SELECT signal from Floppy Disk Controller Module
4	X	—	Select soft sector
5	X	—	Power to stepper motor is always applied
6	—	X	READY signal controlled by SELECT signal from Floppy Disk Controller Module

TABLE 3-4. Jumper Plug Connections

JUMPER POSITION	FUNCTION
JPR1	Connects the SELECT 0 signal line from the Floppy Disk Controller Module to the select line on the disk drive when the drive is used in the Drive 0 position.
JPR2	Connects the SELECT 1 signal line from the Floppy Disk Controller Module to the select line on the disk drive when the drive is used in the Drive 1 position.

CHAPTER 4

ILLUSTRATED PARTS BREAKDOWN

4.1 INTRODUCTION

This chapter contains an illustrated parts breakdown that describes and illustrates all replaceable parts for the EXORdisk II Disk Drive Unit. This illustrated parts breakdown consists of Group Assembly Parts List tables and associated figures which illustrate the component parts and their physical relationships. All of the component parts are listed in disassembly order, with each part having an assigned item number. The item number is shown on the accompanying illustration to identify its physical location within the assembly.

4.2 GROUP ASSEMBLY PARTS LIST

The Group Assembly Parts List consists of a complete breakdown of the EXORdisk II into major subassemblies and detailed parts. The major subassemblies consist of the chassis and power supply subassembly and the two disk drive subassemblies. Since identical disk drives are used, only one drive is listed in the parts list and illustrations. Each assembly is then listed in its order of disassembly, immediately followed by its component parts list properly indented to show their relationship to the assembly. The Group Assembly Parts List is divided into five columns of information as follows:

- Figure and Item Number
- Part Number
- Description
- Quantity Per Assembly
- Usage Code

4.2.1 Figure and Item Number

The Figure number column relates that parts list to its associated illustration. The Item Number assigned to each part is shown in the illustration to show its physical location.

4.2.2 Part Number

The Part Number column lists part numbers for all parts that are replaceable in the unit. Items without a part number are listed as No Number. The notation XXX, when used with a part number, indicates various part parameters, such as cable lengths.

4.2.3 Description

The Description column includes designated and descriptive information necessary to adequately define the items. Entries are indented to indicate their relationship to the next higher assembly. The subheading Attaching Parts within this column lists those parts used for attaching the immediately preceding item, unless otherwise indicated.

The symbol ———*——— signifies the end of the attaching parts for the preceding item(s). The symbol NHA refers to the Next Higher Assembly, U/O indicates Used On, NFS indicates Not Furnished Separately, and N/I indicates that the item is Not Illustrated.

4.2.4 Quantity Per Assembly

The Quantity Per Assembly column contains the number of parts required for each assembly or subassembly. The letters A/R denote that the selection of a part, or parts, should be made As Required. REF refers to an assembly or item that is shown completely assembled in a preceding illustration or the item is listed for reference only.

4.2.5 Usage Code

The Usage Code column lists part variations within different models, assemblies, or subassemblies of the same equipment. In cases where the column is blank, the listed part will apply to all models, assemblies, or subassemblies listed in this publication.

Items designed for more than one configuration are coded for usability. Codes used in this publication are as follows:

MODEL	USAGE CODE
142M, 115 Vac, 60 Hz	A
142M, 100/115 Vac, 50 Hz	B
142M, 208/240 Vac, 50 Hz	C

4.3 PURCHASED PARTS

Many parts within the EXORDisk II (such as the disk drives) are obtained directly from a vendor. Motorola exclusive part numbers are indicated as such by the symbol: (M). All parts not so indicated may be obtained directly from California Computer Products Inc. (CalComp) or other vendors.

4.4 ACCESSORIES

The following accessories are available from CalComp for use with the Model 142M Floppy Disk Drive:

P/N 13810-001	Cover (PWB Side)
P/N 13810-002	Cover (Impeller Side)
P/N 14066-001	Model 2140 Disk Drive Exerciser
P/N 17037-001	Alignment Diskette

4.5 DISK DRIVE PART NUMBER DESIGNATIONS

The disk drive model and options used in the disk drive are indicated by the part number located on the left-hand side of the disk drive. If, for any reason, the disk drive manufacturer is contacted directly, this part number must be used. The following list provides the part number of each model disk drive used in the EXORDisk II.

P/N 17652-011	Model 142M disk drive, 115 Vac, 60 Hz with write protection and gray front panel and handle.
P/N 17652-014	Model 142M disk drive, 208/240 Vac, 50 Hz with write protection and gray front panel and handle.
P/N 17652-013	Model 142M disk drive, 100 Vac, 50 Hz with write protection and gray front panel and handle.
P/N 17652-111	Model 142 disk drive, 115 Vac, 60 Hz with write protection and black front panel and handle.
P/N 17652-114	Model 142M disk drive, 208/240 Vac, 50 Hz with write protection and black front panel and handle.
P/N 17652-113	Model 142M disk drive, 100 Vac, 50 Hz with write protection and black front panel and handle.

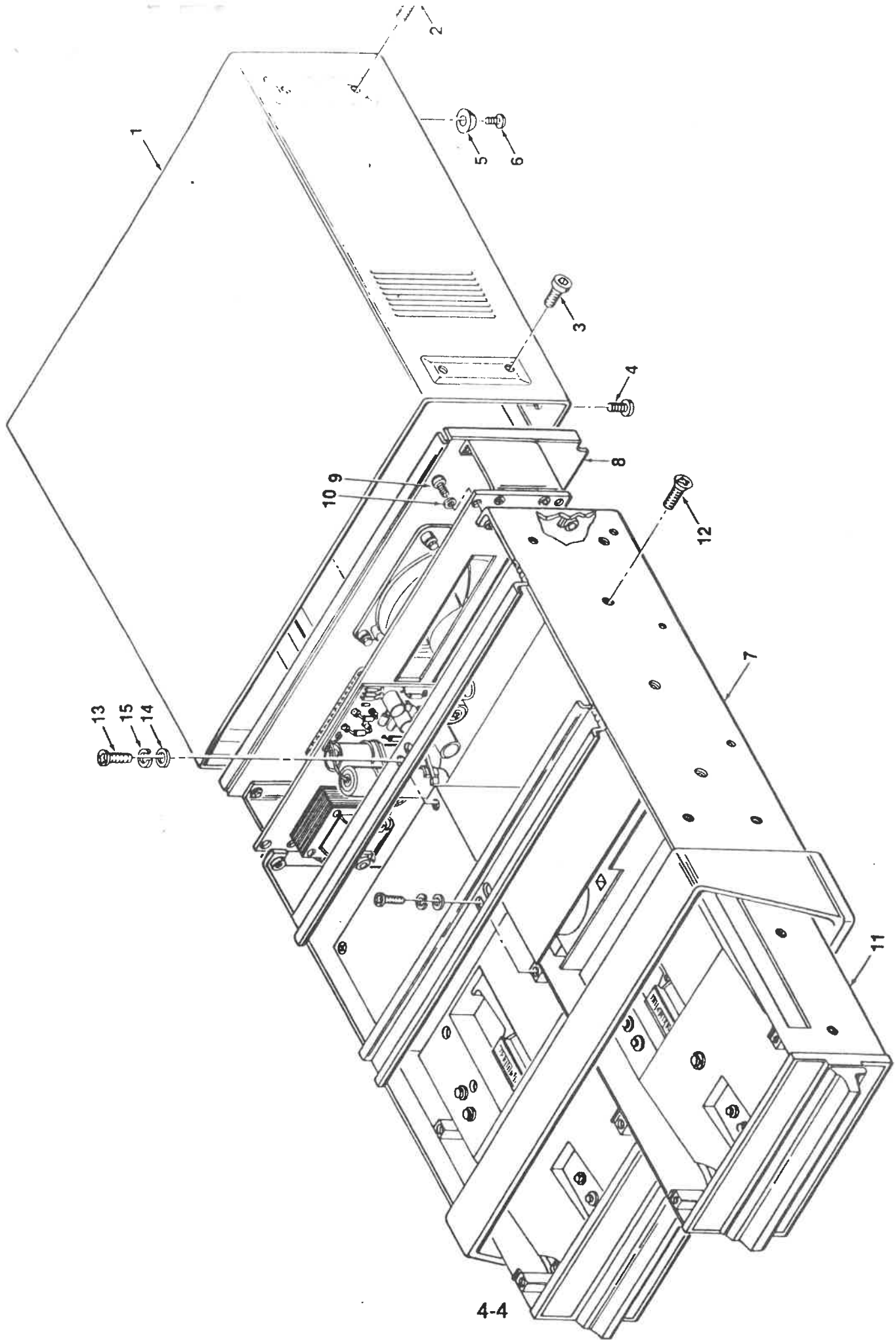
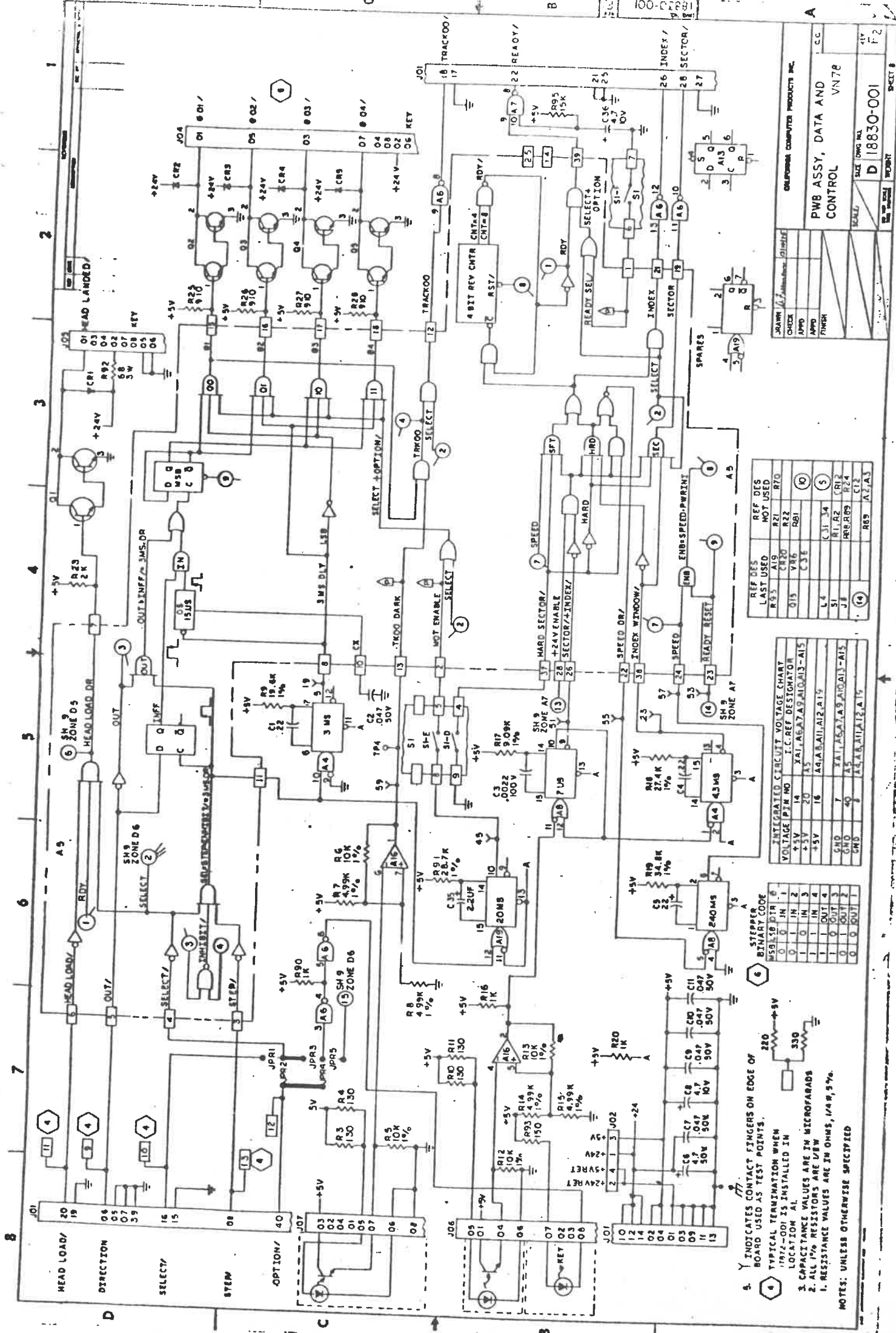


FIGURE 4-1. EXORDisk II Chassis Assembly



4. Y INDICATES CONTACT FINGERS ON EDGE OF BOARD USED AS TEST POINTS.

5. TYPICAL TERMINATION WHEN 1992-001 IS INSTALLED IN LOCATION A1

6. CAPACITANCE VALUES ARE IN MICROFARADS

7. ALL 1% RESISTORS ARE 1/8W

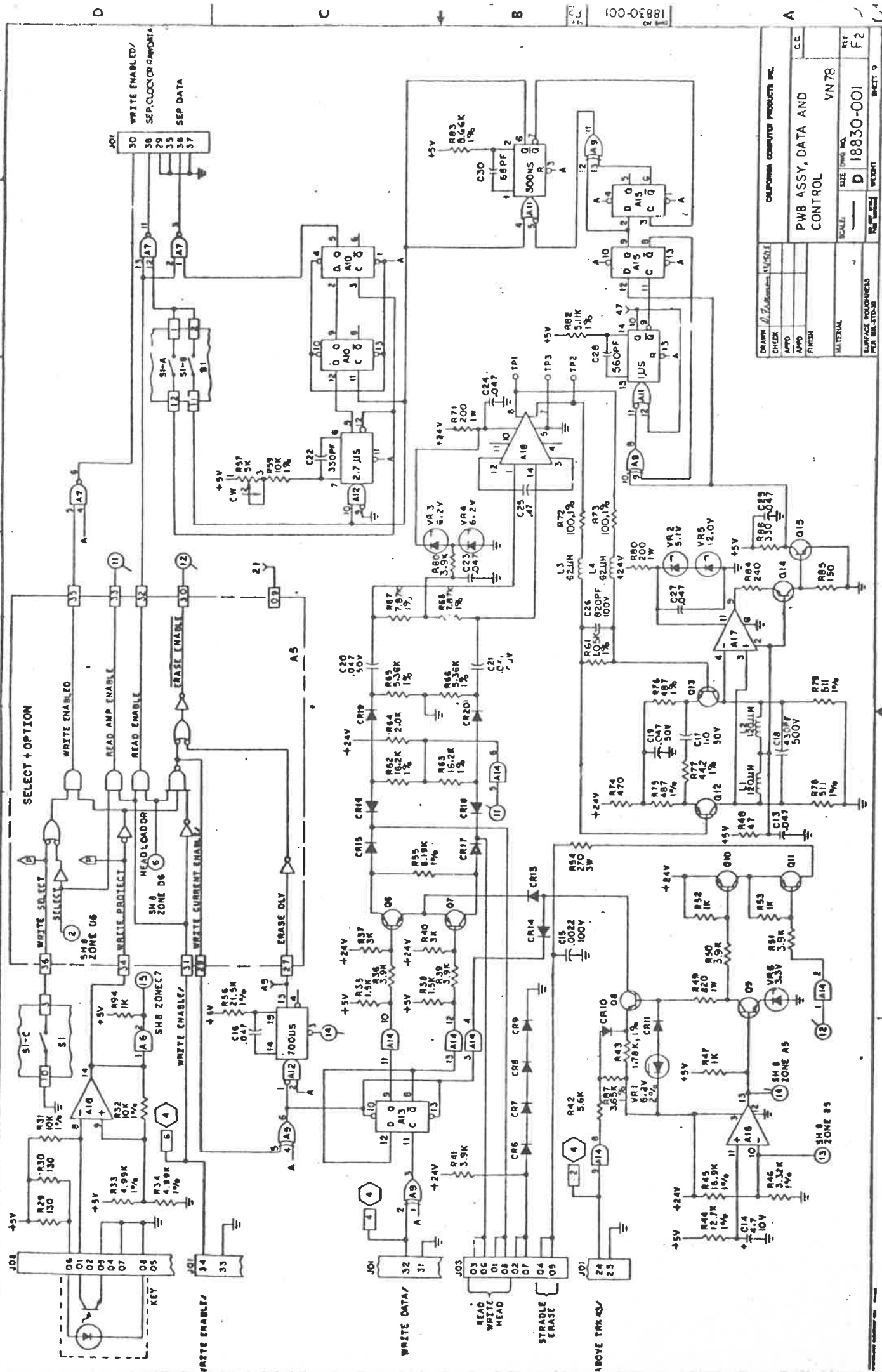
8. RESISTANCE VALUES ARE IN OHMS, 1/8W, 5%.

NOTES: UNLESS OTHERWISE SPECIFIED

INTEGRATED CIRCUIT VOLTAGE CHART	
VOLTAGE (PTM NO)	I.C. REF DESIGNATOR
+5V	14
+5V	15
+5V	20
+5V	16
CHD	7
CHD	15
CHD	2

REF DES	LAST USED	REF DES NOT USED
R95	A19	R21 R70
	C20	R22
Q15	V46	R81
	C3E	(K)
L4		(S)
J1	C3 J4	R1, R2 R12
SF		R8, R9 R24
(14)		C12 C13
		D8C A3A

JOANNE	Address	12111111
CHECK		
AMP		
AMP		
FLUOR		



MCS6501-MCS6505 MICROPROCESSOR INSTRUCTION SET - ALPHABETIC SEQUENCE

ADC	Add Memory to Accumulator with Carry	JSR	Jump to New Location Saving Return Address
AND	"AND" Memory with Accumulator	LDA	Load Accumulator with Memory
ASL	Shift Left One Bit (Memory or Accumulator)	LDX	Load Index X with Memory
BCC	Branch on Carry Clear	LDY	Load Index Y with Memory
BCS	Branch on Carry Set	LSR	Shift Right One Bit (Memory or Accumulator)
BEQ	Branch on Result Zero	NOP	No Operation
BIT	Test Bits in Memory with Accumulator	ORA	"OR" Memory with Accumulator
BMI	Branch on Result Minus	PHA	Push Accumulator on Stack
BNE	Branch on Result not Zero	PHP	Push Processor Status on Stack
BPL	Branch on Result Plus	PLA	Pull Accumulator from Stack
BRK	Force Break	PLP	Pull Processor Status from Stack
BVC	Branch on Overflow Clear	ROL	Rotate One Bit Left (Memory or Accumulator)
BVS	Branch on Overflow Set	ROR	Rotate One Bit Right (Memory or Accumulator)
CLC	Clear Carry Flag	RTI	Return from Interrupt
CLD	Clear Decimal Mode	RTS	Return from Subroutine
CLI	Clear Interrupt Disable Bit	SBC	Subtract Memory from Accumulator with Borrow
CLV	Clear Overflow Flag	SEC	Set Carry Flag
CMP	Compare Memory and Accumulator	SED	Set Decimal Mode
CPX	Compare Memory and Index X	SEI	Set Interrupt Disable Status
CPY	Compare Memory and Index Y	STA	Store Accumulator in Memory
DEC	Decrement Memory by One	STX	Store Index X in Memory
DEX	Decrement Index X by One	STY	Store Index Y in Memory
DEY	Decrement Index Y by One	TAX	Transfer Accumulator to Index X
EOR	"Exclusive-Or" Memory with Accumulator	TAY	Transfer Accumulator to Index Y
INC	Increment Memory by One	TSX	Transfer Stack Pointer to Index X
INX	Increment Index X by One	TXA	Transfer Index X to Accumulator
INY	Increment Index Y by One	TXS	Transfer Index X to Stack Pointer
JMP	Jump to New Location	TYA	Transfer Index Y to Accumulator